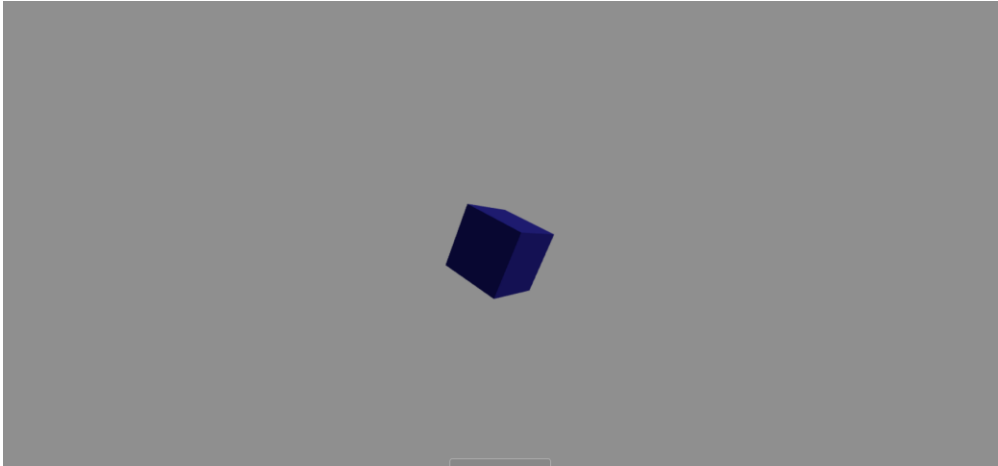


AR/VR ASSIGNMENT-1

The goal of this project was to demonstrate 3 Degrees of Freedom (3DoF) rotation across the X, Y, and Z axes using an object. I choose a cube

Task 1:

I created a simple 3d environment with a camera and a blue cube with three.js . To show the three degrees of rotation, I wrote an animation loop that adds a small, different value to the cube's rotation.x, rotation.y, and rotation.z properties on every single frame. And also, I enabled WebXR to communicate with VR headset(If I ever get a headset)



Task 2:

For the AR work, I used A-Frame and ar.js. The code uses the device's webcam to scan the environment. The moment the math detects the "Hero" pattern, it calculates exactly where to place the 3D cube so it looks like it's sitting physically on top of the screen. For animatio, I used A-Frame's animation component to rotate the object from 0 to 360 degrees.

